

Graph-Based Cluster Dynamics Modeling of Ferritic–Martensitic Steels and Zirconium Alloys

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1 Graph-Based Cluster Dynamics

polarity, the reaction admissibility graph, the graph walker

2 Mean-Field Irradiated Ferritic–Martensitic Steels

EUROFER-97: conservation, growth windows, `RADCLUSTER`

3 Beyond Mean-Field: Spatially Resolved CD

what a field buys that a number cannot

4 Irradiated Zirconium

nine crystallographic families, self-consistent bias

5 Conclusions

Why Graph Cluster Dynamics (CD)?

Graph is the interface

one reaction graph,
not one material's
hard-coded reaction list

Code is inherited

solver, preconditioner,
reductions & diagnostics
reused, not rewritten

Hypergraph kinetics

species = vertices,
reactions = hyperedges;
rates read off a
stoichiometric matrix

Topology fixes invariants

conservation laws &
steady states set by
network structure alone
(Horn–Jackson, Feinberg)

No double counting

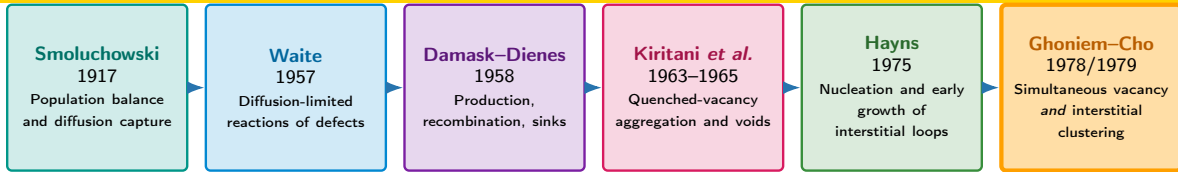
walker visits each
admissible edge *once*; no
hand-written balances

Conserved by construction

Frenkel-pair & He balances
monitored, not imposed

bcc Fe (EUROFER-97) $\rightarrow$ hcp Zr
same core, two host declarations

Cluster Dynamics Lineage: Key ideas during a century of research



The capture kernel came first; the *coupled* evolution of both defect-cluster families under continuous irradiation came later.

$$K_{ij} = 4\pi (D_i + D_j) R_{ij} \quad \Longrightarrow \quad \{C_n^v(t), C_n^i(t)\} \text{ under a continuous source}$$

An early systematic irradiation cluster-dynamics model — and plausibly the first fully coupled treatment of simultaneous vacancy and interstitial clustering under continuous irradiation.

- [1] Smoluchowski, *Z. Phys. Chem.* 92, 129 (1917). [2] Waite, *Phys. Rev.* 107, 463 (1957). [3] Dienes & Damask, *J. Appl. Phys.* 29, 1713 (1958). [4] Hayns, *The nucleation and early growth of interstitial dislocation loops in irradiated materials*, *J. Nucl. Mater.* (1975).
[5] Ghoniem & Cho, UCLA-ENG-7845 (1978). [6] Ghoniem & Cho, *phys. status solidi (a)* 54, 171 (1979).

1. Graph-Based Cluster Dynamics

The prediction problem

Operating envelope of fission and fusion structures

- ▶ Displacement rates 10^{-7} – 10^{-6} dpa s⁻¹, accumulating 10–200 dpa
- ▶ Loops, voids, bubbles, segregation, precipitation $\Rightarrow$ hardening, embrittlement, swelling, creep, DBTT shift

Cluster dynamics is the workhorse mesoscale theory

- ▶ Time scales 10^{-6} – 10^8 s, far beyond MD
- ▶ Explicit cluster-size distribution $\Rightarrow$ every quantity a property model needs
- ▶ Cost low enough to be the meeting point of atomistic parameterization and engineering prediction

Four persistent gaps

- 1 **No host independence at the algorithmic core** — a new material means rewriting the solver, though the mathematics is unchanged.
- 2 **He–V state space** — resolving $\text{He}_\ell V_m$ over the whole (ℓ, m) grid is prohibitive; mean-loading reductions are used heuristically, without stated limits or preserved moments.
- 3 **Size-distribution closure** — grouping schemes leak mass unless the inter-bin flux is designed; closure error is seldom quantified.
- 4 **Stiffness and solver design** — the Jacobian is banded and bordered; generic preconditioners perform poorly, and conservation is rarely diagnosed.

Ghoniem, *A Graph-Based Cluster Dynamics Framework for Irradiated Materials* (2026), §1

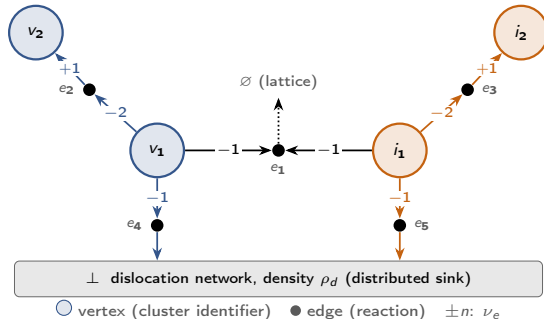
The Reaction Admissability Graph (RAG)

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \nu, k)$$

- $\mathcal{V}$: admissible cluster identifiers
- $\mathcal{E}$: elementary reactions, each carrying a stoichiometric tuple ν_e
- $k : \mathcal{E} \rightarrow \mathbb{R}_{\geq 0}$, rate kernel $k_e(t, s, \Theta)$

Host-specific in $(\mathcal{V}, \mathcal{E}, \nu)$, **calibration-specific** in k — the algorithmic core is the **graph walker**.

Example: monomers, di-clusters, and a dislocation sink



The ten edge classes — every edge lies in exactly one

1	Growth	$(\sigma, n) \rightarrow (\sigma, n+1)$, σ -monomer capture
2	Shrinkage	$(\sigma, n) \rightarrow (\sigma, n-1)$, $-\sigma$ impingement
3	Dissociation	thermal emission; binding energy
4	Recombination	$-\sigma$ monomer annihilated, not pooled
5	Annihilation	binary, cross-polarity; quadratic kernel
6	Inter-population	$p \rightarrow p'$ at fixed (σ, n, c)
7	Solute trap/detrap	$c \rightarrow c \pm e_s$
8	Coalescence	$(\sigma, n) + (\sigma, n') \rightarrow (\sigma, n+n')$
9	Source	$\emptyset \rightarrow \tau$; cascade spectrum
10	Sink	$\tau \rightarrow \emptyset$; surface, GB, dislocation

Annihilation (5): the larger cluster survives, shrunk by what it absorbed.

Example edges — ν_e listed on (v_1, i_1, v_2, i_2)

edge	reaction	class	ν_e	kernel k_e
e_1	$v_1 + i_1 \rightarrow \emptyset$	4	$(-1, -1, 0, 0)$	$4\pi(D_v + D_i)R_{vi}$
e_2	$v_1 + v_1 \rightarrow v_2$	1	$(-2, 0, +1, 0)$	$4\pi D_v R_{vv}$
e_3	$i_1 + i_1 \rightarrow i_2$	1	$(0, -2, 0, +1)$	$4\pi D_i R_{ii}$
e_4	$v_1 \rightarrow \perp$	10	$(-1, 0, 0, 0)$	$Z_v \rho_d D_v$
e_5	$i_1 \rightarrow \perp$	10	$(0, -1, 0, 0)$	$Z_i \rho_d D_i$

The walker reads each edge once: $J_e = k_e \prod_{a \in \text{in}(e)} c_a$, $\dot{c}_a = \sum_e \nu_{a,e} J_e$

e.g. $\dot{c}_{v_1} = -k_{e_1} c_{v_1} c_{i_1} - 2k_{e_2} c_{v_1}^2 - k_{e_4} c_{v_1}$

The master equation is a graph walk

$$\dot{\mathbf{c}} = \underbrace{\mathbf{G}}_{\text{cascade source}} + \underbrace{\mathbf{S}\mathbf{J}(\mathbf{c})}_{\text{walk over } \mathcal{E}} - \underbrace{\mathbf{D}\mathbf{c}}_{\text{unresolved fixed sinks}}, \quad \mathbf{D} = \text{diag}(\mathcal{D}_1, \dots, \mathcal{D}_{|\mathcal{V}|})$$

Read per vertex, with the summation convention over the admissible edge set $\mathcal{E}$: $\dot{c}_v = G_v + S_{ve}J_e(c) - \mathcal{D}_v c_v$.

- ▶ $\mathbf{c} \in \mathbb{R}_{\geq 0}^{|\mathcal{V}|}$ — the **concentration vector**: one component per vertex $v \in \mathcal{V}$, a cluster population or a matrix gas reservoir — *not* a physical size
- ▶ $\mathbf{S} \in \mathbb{Z}^{|\mathcal{V}| \times |\mathcal{E}|}$ — the **stoichiometric matrix**: column e is that edge's update vector $\boldsymbol{\nu}_e$, and $S_{ve} = \nu_{e,v}$ is the net number of v produced by one occurrence of e ; negative when v is consumed
- ▶ $\mathbf{J}(\mathbf{c}) \in \mathbb{R}_{\geq 0}^{|\mathcal{E}|}$ — the **flux vector**: one *scalar* per edge, $J_e = k_e \prod_v c_v^{m_e^T(v)}$, i.e. G_e (source), $k_e c_v$ (unary), $k_e c_v c_w$ or $k_e c_v^2$ (binary)

k_e is the **kernel** the edge carries — the state-dependent function; the value it takes once evaluated is the **rate coefficient** (written $\mathcal{K}$ in the reaction table that follows). Concentrations are *atomic fractions*, so $[k_e] = [G_e] = [J_e] = \text{s}^{-1}$ (**Hz**): every kernel is a frequency, the volumetric coefficient being $k_e^{\text{vol}} = \Omega k_e$ with $\Omega = a^3/2$.

The network fixes $\mathbf{S}$, the calibration fixes $\mathbf{J}$, and nothing else enters. The right-hand side is *assembled*, never written down: loop over admissible edges, evaluate the flux, accumulate the stoichiometric contribution.

Reaction content of the EUROFER-97 graph

Three rows of the table — class, edge, stoichiometry, flux

Reaction class	Edge e	Non-zero entries of $\nu_e = S_e$	Flux J_e
(8) Coalescence (SIA–SIA), P3	$I_{n'} + I_{n''} \rightarrow I_{n'+n''}$	$+1_{n'+n''} - 1_{n'} - 1_{n''}$ (SIA block)	$\frac{1}{2} \mathcal{K}_{n',n'',c_{n'}c_{n''}}^{ii}$
(8) Coalescence (vacancy–vacancy), P2v	$V_{m',0} + V_{m,\ell} \rightarrow V_{m+m',\ell}$	$+1_{m+m',\ell} - 1_{m',0} - 1_{m,\ell}$ (cavity block)	$\frac{1}{2} \mathcal{K}_{m',m,c_{m',0}c_{m,\ell}}^{vv}$
(5) Annihilation (SIA–vacancy), P1, P2i	$I_n + V_{m,\ell} \rightarrow \text{survivor}$	$-1_n - 1_{m,\ell}$, plus $+1_{n-m}$ if $n > m$ or $+1_{m-n,\ell}$ if $m > n$	$\mathcal{K}_{n,m}^{ic} c_n c_{m,\ell}$

Three of twelve rows. The full table carries one row per admissible edge family: (3) dissociation — thermal SIA, vacancy and He emission; (5) annihilation; (6) inter-population transfer; (7) composition — gas trapping, thermal gas emission, re-resolution; (8) coalescence; (9) cascade source G; (10) loss to unresolved sinks Dc; (11) Frenkel-pair emission (trap mutation). Growth and shrinkage carry no rows of their own — they are the $n' = 1$ and the $n > m$ branches above.

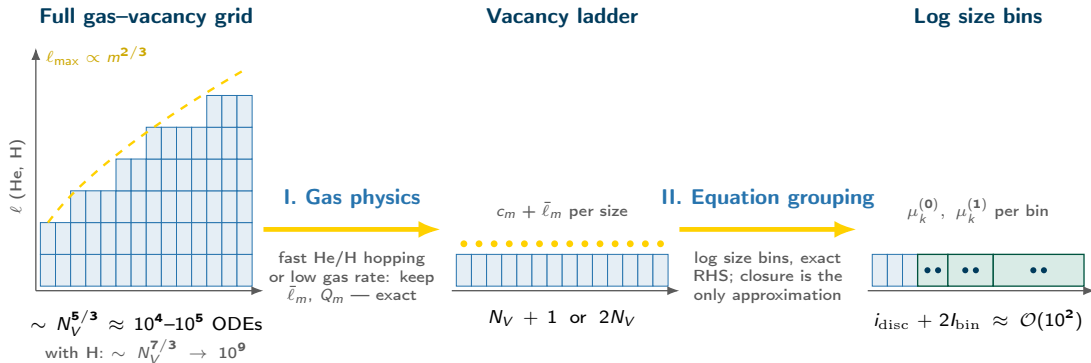
The same row, read at $n' = n'' = 1$: the two edges of the RAG example

- ▶ **Di-interstitial** $I_1 + I_1 \rightarrow I_2$: $\nu_e = +1_2 - 21_1$, $J_e = \frac{1}{2} \mathcal{K}_{1,1}^{ii} c_1^2 \Rightarrow +\frac{1}{2} \mathcal{K}_{1,1}^{ii} c_1^2$ into $\dot{c}_2$ and $-\mathcal{K}_{1,1}^{ii} c_1^2$ into $\dot{c}_1$
- ▶ **Di-vacancy** $V_1 + V_1 \rightarrow V_2$: $\nu_e = +1_2 - 21_1$, $J_e = \frac{1}{2} \mathcal{K}_{1,1}^{vv} c_1^2 \Rightarrow$ the same two entries with $\mathcal{K}_{1,1}^{vv}$
- ▶ **Di-vacancy in components** on the ordered vertex list (v_1, i_1, v_2, i_2) of the RAG: $\nu_e = [-2, 0, +1, 0]^T$ — two v_1 consumed, one v_2 produced, both SIA entries zero

The $\frac{1}{2}$ removes the double count of the ordered reactant pair; the walker visits the edge *once* and the -2 in ν_e does the rest — no hand-written balance, and no double counting.

Collapsing the state space: gas physics & equation grouping

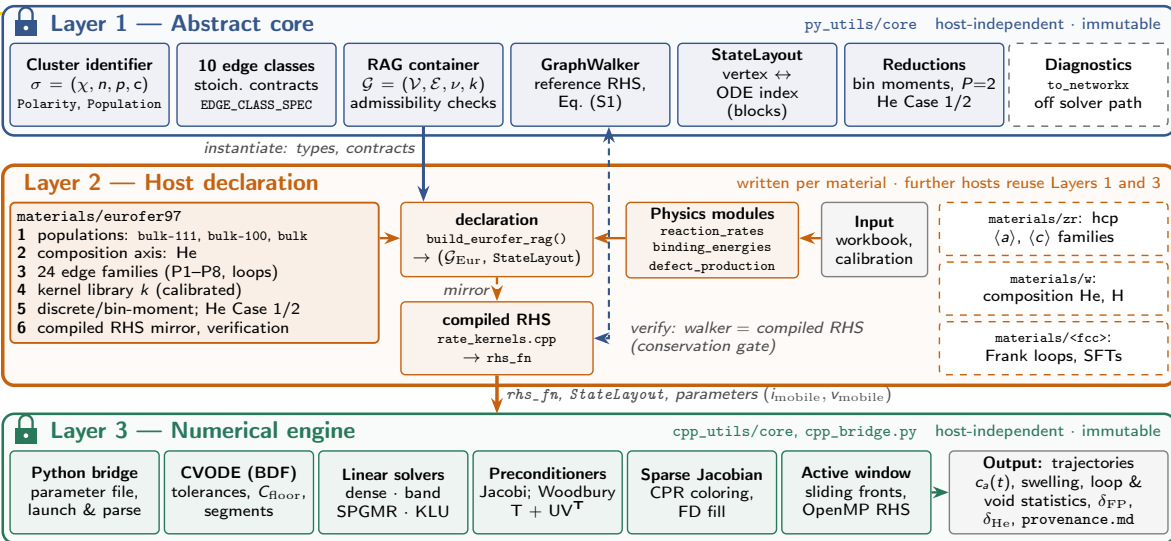
Observed ladders reach $N_I, N_V \sim 10^3\text{--}10^4 \Rightarrow$ the full (m, ℓ) grid is too large and too stiff to integrate.



$\mathcal{O}(N_V^{5/3}) \rightarrow \mathcal{O}(\log N)$ **unknowns** — reduction is a prerequisite, not a refinement.

grouping lineage: Kiritani 1973; Golubov 2001; Jourdan 2014; Kohnert 2017; Terrier 2017

RadCluster architecture: three layers



2. Irradiated Ferritic–Martensitic Steels: The Mean-Field Limit

instantiating the graph for EUROFER-97

Instantiating the graph: EUROFER-97

Three declarations, no solver changes

- ▶ Gas list $N_g = 1$, $g_1 = \text{He}$, so $c \rightarrow \ell \in \mathbb{Z}_{\geq 0}$ — *helium is the only resolved trapped solute*
- ▶ One population per polarity, $\mathcal{P}_{\pm}^{\text{Eur}} = \{\text{bulk}\}$
- ▶ Saturation $\ell_{\max}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor$, $\alpha_{\text{He}} \in [1.5, 2.0]$

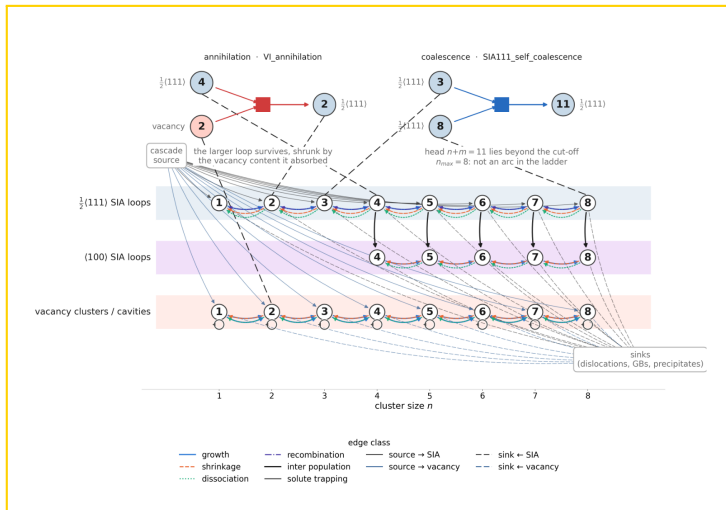
Three tracked populations

- ▶ $c_n(t)$ — SIA clusters; 3D-mobile for $n=1, 2, 3$, $\frac{1}{2}\langle 111 \rangle$ loops for $n \geq 4$; carry no He. *The distinction is a size-dependent kernel, not a population label*
- ▶ $c_{m,\ell}(t)$ — cavities: voids ($\ell = 0$), He bubbles ($\ell \geq 1$)
- ▶ $c_h(t)$ — free helium

GBs, the dislocation network and MX / M_{23}C_6 precipitates are *unresolved fixed sinks*.

The graph is generated, not drawn

- ▶ The RAG at right is *materialized* from these declarations and drawn straight from the graph object by open-source code, github.com/Ghoniem-Org/RadCluster — no edge list is hand-transcribed, so the figure cannot drift from the model



Conservation laws are a consequence of stoichiometry

Frenkel-pair law — the swelling identity

(exact)

$$S - S_I - \Delta J_i^{\text{fix}} + \Delta J_v^{\text{fix}} = \text{const} \Rightarrow \boxed{S = S_I + \Delta J_i^{\text{fix}} - \Delta J_v^{\text{fix}}}$$

- ▶ $S = \sum_m m \bar{c}_m$: vacancies stored in cavities (swelling)
- ▶ $S_I = \sum_n n c_n$: interstitials in SIA clusters and loops
- ▶ $\Delta J_{i,v}^{\text{fix}}$: cumulative SIA / vacancy flux into fixed sinks

Helium law

$$\frac{dS_{\text{He}}}{dt} = G_{\text{He}} - k_h^2 D_h c_h - k_v^2 D_v \sum_{m \leq m_{\text{max}}^v} \sum_{\ell} \ell c_{m,\ell}$$

- ▶ S_{He} : He in all resolved clusters; G_{He} : He production
- ▶ $k_h^2 D_h c_h$: mobile interstitial He lost to fixed sinks
- ▶ last term: mobile He–vacancy clusters (m, ℓ) absorbed at fixed sinks, each carrying ℓ He

Why they follow from stoichiometry

Edge e carries a stoichiometric vector ν_e , a column of the stoichiometric matrix $\mathbf{N}$. A conserved quantity is a **left null vector $\mathbf{w}$ of $\mathbf{N}$ restricted to internal edges**, $\mathbf{w}^T \nu_e = 0$; its entry w_a is the content of vertex a : $w_a = +m$ for v_m , $-n$ for i_n (Frenkel pair), or $w_a = \ell$ (He). Then

$$\frac{d}{dt} \mathbf{w}^T \mathbf{c} = \sum_e (\mathbf{w}^T \nu_e) J_e,$$

internal edges cancel identically and only fixed-sink edges (plus the He source) survive — **for any kernel values**.

What they buy for modeling

- ▶ **Verification**: $\delta_{\text{FP}}, \delta_{\text{He}} \lesssim 10^{-6}$; larger flags a bug, missing reaction, or loose tolerance
- ▶ **Calibration-independent**
- ▶ **Insight**: swelling from SIA bookkeeping
- ▶ **Safe reductions**: groupings must preserve $\mathbf{w}^T \mathbf{c}$; monitored, not imposed

Example (RAG of the earlier slide), Frenkel-pair law: $\mathbf{w} = (+1, -1, +2, -2)$ on (v_1, i_1, v_2, i_2)

e_1 : $v_1 + i_1 \rightarrow \emptyset$: $(+1)(-1) + (-1)(-1) = 0$ e_2 : $v_1 + v_1 \rightarrow v_2$: $(+1)(-2) + (+2)(+1) = 0$ e_3 : $i_1 + i_1 \rightarrow i_2$: $(-1)(-2) + (-2)(+1) = 0$

source $\emptyset \rightarrow v_1 + i_1$: $(+1) + (-1) = 0$ e_4 : $v_1 \rightarrow \perp$: $(+1)(-1) = -1$ e_5 : $i_1 \rightarrow \perp$: $(-1)(-1) = +1$

$$\Rightarrow \frac{d}{dt} (S - S_I) = J_{e_5} - J_{e_4} = \frac{d}{dt} (\Delta J_i^{\text{fix}} - \Delta J_v^{\text{fix}})$$

Cavity and Dislocation Loop Co-Growth Conditions

Mobile inventories and their ratio,

$$\Phi_I = \sum_{n \leq i_{\text{mob}}} n D_n c_n, \quad \Phi_V = \sum_{m \leq v_{\text{mob}}} m D_m c_{m,0}, \quad x \equiv \Phi_V / \Phi_I.$$

(a) Elastic bias alone

$$\underbrace{1 + \varepsilon_{m,\ell}}_{\text{cavity grows above}} < x < \underbrace{Z_i^{\text{loop}}}_{\text{loops grow below}}$$

The critical-radius statement in flux-ratio form; emission closes it from below, helium widens it by lowering $\varepsilon_{m,\ell} = D_v c_v^{\text{th}} / \Phi_I$.

(b) Elastic + production bias

$$1 + \varepsilon_{m,\ell} - \underbrace{\gamma_v}_{\text{prod.}} < x < \underbrace{Z_i^{\text{loop}}}_{\text{elastic}} + \underbrace{\gamma_i}_{\text{production}}$$

$$\gamma_i = \frac{f_i^{\text{cl}}}{1 - f_i^{\text{cl}}} \frac{k_I^2}{k_L^2}, \quad \gamma_v = \frac{f_v^{\text{cl}}}{1 - f_v^{\text{cl}}} \frac{k_V^2}{k_C^2}$$

Clustered SIAs are deposited *directly* into loops (k_L^2) and clustered vacancies into cavities (k_C^2), independent of x ; with $k_I^2 \Phi_I \simeq G(1 - f_i^{\text{cl}})$ this adds γ_i to the loop edge.

α -Fe cascade statistics

	fission	fusion
mono-SIA fraction $1 - f_i^{\text{cl}}$	0.42	0.35
mono-vacancy fraction $1 - f_v^{\text{cl}}$	0.85	0.80
mean SIA cluster size	4.4	6.3
mean vacancy cluster size	2.6	3.1
largest SIA cluster	20	50
largest vacancy cluster	10	20
He co-production (appm/dpa)	0.5–1	~10

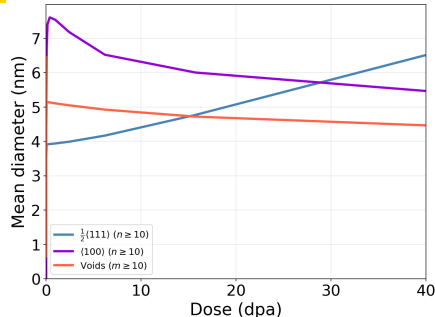
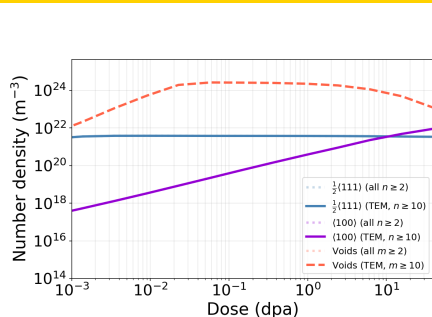
(a) Iron misses the elastic window

$x \simeq \eta k_I^2 / k_V^2$, $\eta = (1 - f_v^{\text{cl}}) / (1 - f_i^{\text{cl}}) \simeq 2.0\text{--}2.3$, but Z_i^{loop} exceeds unity by only a few percent ($f_i^{\text{cl}} - f_v^{\text{cl}} \lesssim 0.04$ needed; cascades give **0.43–0.45**).

(b) Production bias opens it

$\gamma_i \gtrsim f_i^{\text{cl}} / (1 - f_i^{\text{cl}}) \approx 1.4$ (fission), 1.9 (fusion), so $Z_i^{\text{loop}} + \gamma_i \gtrsim 2.4\text{--}2.9 > x$: **co-growth is a production-bias phenomenon**.

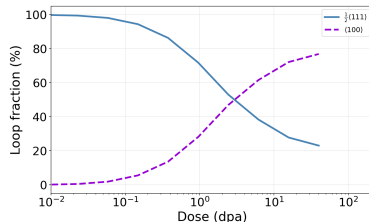
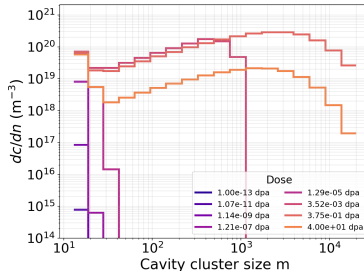
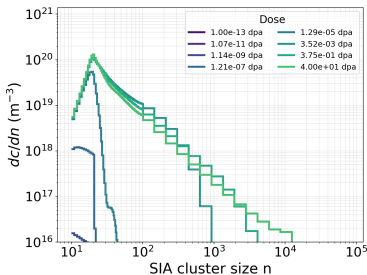
EUROFER-97 to 100 dpa: what the run produces



Left: number densities of SIA loops and of cavities, counted inside the TEM-visible window. Right: the corresponding mean sizes. Fission spectrum, 300°C, 10^{-6} –100 dpa; $I_{\max} = 8 \times 10^4$, $V_{\max} = 2 \times 10^4$, $i_{\text{mob}} = v_{\text{mob}} = 5$.

- ▶ **Incubation → rapid nucleation → saturation:** the long nucleation transient is a direct consequence of *continuous coalescence of 1D-glissile SIA clusters* — a channel a mono-defect treatment cannot reproduce
- ▶ **<111> size rises steadily:** once their density saturates, glissile loops **coarsen at nearly fixed number** — by coalescence with 1D-migrating SIA clusters and direct production-bias deposition, neither dependent on x
- ▶ **<100> rises, then falls:** the first sessile loops, formed by <111>–<111> junction reactions, are few and capture a large share of the SIA flux; as <111> density builds, new small <100> loops keep forming and dilute the mean, while the largest are absorbed into the network
- ▶ **Void density rises then falls; mean size drifts down:** He-stabilized nuclei fill the visible window early; later **1D-migrating <111> SIA clusters** strike cavities, each delivering n SIAs at once — small cavities ($m \lesssim n$) are annihilated and larger ones shrink, a loss the mono-vacancy supply cannot offset

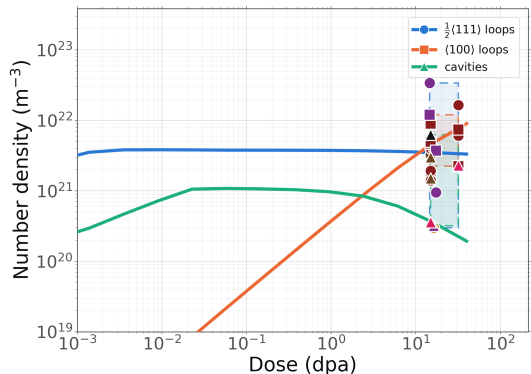
Behind the averages: distributions and loop character



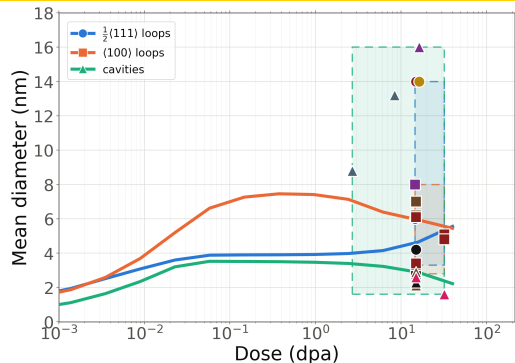
SIA and vacancy cluster distributions on a TEM size axis at several doses, and the fraction of the SIA inventory held in loops.

- ▶ The SIA distribution stays **narrow** and marches rapidly to large sizes; the vacancy distribution is **broader and slower**, with thermal emission shaping the high-size tail — two shapes no single mean size can stand in for
- ▶ Most of the SIA inventory sits in clusters *below* the TEM detection limit at low dose: the visible density is a **window onto the distribution**, not the distribution
- ▶ The loop fraction rises and then saturates — the signature of **direct production of loops by G_n** rather than absorption

Against experiment: density and size versus dose



●▲ Dethloff 2016 ● Klimenkov 2011 ▲ Weiß 2012 ●▲ Chauhan 2021
 ● Dethloff 2018 ●▲ Klimenkov 2020



●▲ Dethloff 2016 ● Klimenkov 2011 ▲ Weiß 2012 ▲ Coppola 2019
 ● Dethloff 2018 ●▲ Klimenkov 2020 ●▲ Chauhan 2021

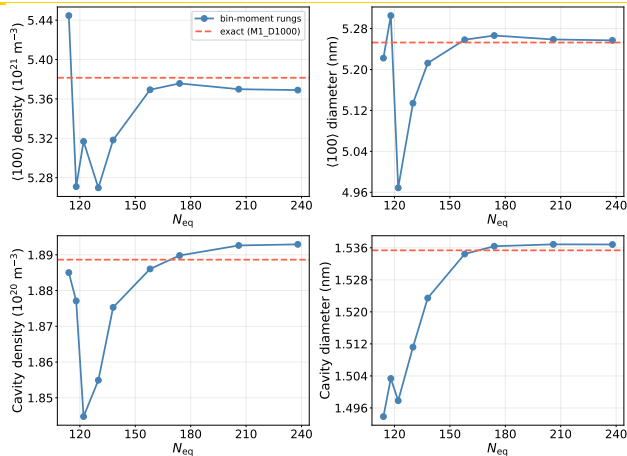
- ▶ Computed densities fall within the same order of magnitude as the TEM data and the mean sizes agree **without parameter tuning**
- ▶ The experimental scatter between campaigns is itself a **factor of a few** — comparison is meaningful only at that level, and only inside the stated visibility cut-off

Data sources

Weiß, Gaganidze & Aktaa, *J. Nucl. Mater.* 426 (2012) — EUROFER97, ARBOR (BOR-60), up to 32 dpa

Klimenkov, Materna-Morris & Möslang, *J. Nucl. Mater.* 417 (2011) 124 — SPICE (HFR), 16.3 dpa, 250–450°C; $\frac{1}{2}\langle 111 \rangle$ loops and He bubbles

Dethloff, Gaganidze & Aktaa, *Nucl. Mater. Energy* 15 (2018) 23 —



$\langle 100 \rangle$ and cavity observables against N_{eq} at a fixed discrete core $i_{disc} = v_{disc} = 50$, $I = V = 1000$, 20 dpa; the exact arm dashed. Each closes on the reference by thirteen bins, at 158 equations.

- ▶ Sweeping *only* the bin count: **five of the six observables converge, and converge quickly** — by thirteen bins, 158 equations against 2006 for the exact arm, N_{100} , d_{100} , N_{cav} and d_{cav} all lie within 0.3%; N_{111} plateaus at a residual +1.0%
- ▶ **The sixth does not:** d_{111} stays between -5 and -6.6% across nine runs, and the finest rung is the worst of them
- ▶ Sweeping the two axes independently identifies the governing one: the error falls down the **discrete-core** column (-6.3, -3.4, -2.0, -0.1, +0.7% at $I_{bin} = 8$) and is nearly *flat* across the bin count; at $i_{disc} = 800$ it is better than 1% at every bin count
- ▶ Why: the $\frac{1}{2} \langle 111 \rangle$ mean size is ≈ 147 interstitials, about three times the discrete core, so the peak of its distribution lies in the binned region and is *carried by the closure rather than resolved*

Three alternatives excluded by direct measurement

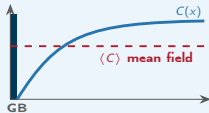
Code path — the bin-moment path run without closure ($I_{bin} = 0$, $i_{disc} = I$) reproduces the discrete arm at 20 dpa to within rounding. **Domain edge** — the exact spectrum decays smoothly to the top of the grid, which holds 0.01% of the content. **Slow accumulation** — the deficit already stands at -8.4% at 0.005 dpa and drifts only to -6.6% by 20 dpa.

3. Beyond Mean-Field: The Spatially-Resolved Cluster Dynamics (SRCD)

Three assumptions are folded into a mean field — each fails

Classical cluster dynamics is spatially averaged: every concentration is one number for the whole material, and every sink is a k^2 already averaged over a random, uniform arrangement of absorbers.

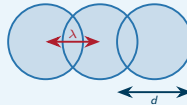
1. Spatial uniformity



A sink strength means something only if the diffusion field relaxes to a common far field before the next absorber. Near a surface, a grain boundary or a denuded band it does not.

Structural, not quantitative: one average then describes *no point* in the material.

2. Diluteness

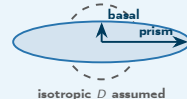


$$d > \lambda$$

Avrami-type kernels assume absorbers are placed independently at random. At 40 dpa basal loops are **195 nm across at 158 nm spacing** — each wider than the distance to its neighbour.

Overlap is then correlated *by construction*, and **the estimate ceases to estimate anything** ($\varphi_{LL} \rightarrow 1$).

3. Isotropy



In α -Zr prism-plane interstitial mobility exceeds basal by about an order of magnitude, yet most treatments keep an isotropic D and recover directionality through *independently fitted* bias factors.

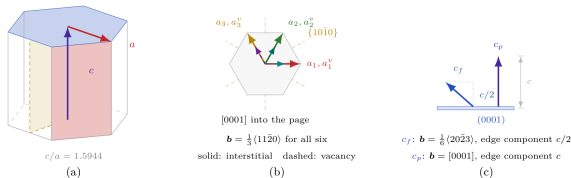
Diffusion is then isotropic and its sinks biased by that same anisotropy — severing atomic parameters from predicted dimensional change.

A spatially averaged model cannot see these failures — it reports the wrong number with no indication that it has.

4. Irradiated Zirconium

a self-consistent, spatially-resolved model for α -Zr

Nine crystallographic families, because a size is not a defect



The hcp cell at the physical axial ratio $c/a = 1.5944$; the three prismatic variants at 60° (solid interstitial, dashed vacancy); the basal pair seen edge-on.

family	habit	b	char.	λ_k (nm)
a_1, a_2, a_3	$\{10\bar{1}0\}$	$\frac{1}{3} \langle 11\bar{2}0 \rangle$	i	0.151
a_1, a_2, a_3	$\{10\bar{1}0\}$	$\frac{1}{3} \langle 11\bar{2}0 \rangle$	v	0.151
c_f faulted	(0001)	$\frac{1}{6} \langle 20\bar{2}3 \rangle$	v	0.170
c_p perfect	(0001)	[0001]	v	0.120
c_0 SF pyramid	(0001) base	partials	v	—

A cluster is not specified by its size alone: its Burgers vector and habit plane determine its contribution to *irradiation growth*, to *hardening*, and to the *sink field* every other species sees.

Nine families = three prismatic $\langle a \rangle$ variants of both characters + a three-state basal vacancy chain $c_0 \rightarrow c_f \rightarrow c_p$. Eight are loops; the stacking-fault pyramid is not.

- ▶ Two families sharing a habit but differing in *character* respond to the same mobile fluxes with opposite sign
- ▶ Two sharing a *character* but differing in habit present differently oriented capture cross-sections to an anisotropic diffusion field — **this is the competition the self-consistent formulation exists to describe without a free parameter**
- ▶ Resolving the three prismatic variants makes the response to an arbitrary stress state a *redistribution among three* rather than a shift of one scalar

Mobile set $\mathcal{M} = \{C_v, C_i, C_{2i}, C_{3i}\}$ — the di- and tri-interstitial are state variables in their own right, each with its own diffusion tensor, anisotropy p_m and capture efficiency. *The three prismatic normals lie in (0001) while the basal normal is [0001]: no single zone axis images both populations at true size.*

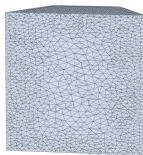
The master equation, and how it is solved in three dimensions

Master equation — the graph walk, with transport

$$\frac{\partial c}{\partial t} = \underbrace{-\nabla \cdot \mathbf{J}(c)}_{\text{transport}} + \underbrace{\mathbf{G}}_{\text{cascade source}} + \underbrace{\mathbf{S}\mathbf{J}(c)}_{\text{walk over } \mathcal{E}} - \underbrace{\mathbf{D}c}_{\text{unresolved fixed sinks}}$$

Sans-serif symbols are vectors and matrices over the vertex (species) set $\mathcal{V}$, one copy per mesh node; *bold italic* denotes Cartesian vectors. $\mathbf{J}_v = -D_v \nabla c_v$ is the Cartesian flux of vertex v , zero on every immobile index, and $c_v = c_v^{\text{eq}}$ on $\partial\Omega_D$ for $v \in \mathcal{M}$. Space adds only the first term: $\mathbf{G}$, $\mathbf{S}$, $\mathbf{J}$ and $\mathbf{D}$ are the mean-field objects, assembled node-wise.

Operator splitting — the spatially coupled mobile solve alternated with node-wise stiff chemistry:



S4: c_M^* frozen $\Rightarrow$
36 stiff ODEs per node, CVODE-BDF

Fast — mobile $c_M(x)$
4 species, 7.6×10^5 dof
spatially coupled

Slow — immobile $Q(x)$
9 × 3 moments,
 5.1×10^6 dof
node-wise, decoupled

S1–S3: Q_n frozen $\Rightarrow$ assemble sinks,
solve $R_M(c_M^*; Q_n) = 0$ by FEM Newton

129 638 tetrahedra, 25 907 vertices, 189 533 CD nodes; 150 nm boundary layer, five elements across.

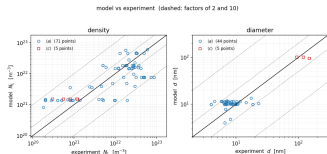
Why operator splitting: $\tau_M \approx \mu\text{s}$ against $\tau_Q \sim 10^7$ s — thirteen decades, so c_M is *slaved* and its solver carries *no* time derivative. **Payoff:** 1.6×10^5 systems of 36 replace one block of 2.8×10^6 — 7.4×10^9 flops against 2.0×10^{20} , 4 h on 40 cores — at the first-order (Lie–Trotter) $O(\Delta\gamma)$ splitting error set by the exchange cadence.

Calibration against 78 TEM measurements: estimator, set, search

The estimator is a maximum *a posteriori* point, not a least-squares fit

$$J(\theta) = \sum_{f \in \{A, C\}} \left[\frac{w_f^N}{|\mathcal{N}_f|} \sum_{r \in \mathcal{N}_f} \left(\log_{10} \frac{N_r(\theta)}{s_f^N N_r^{\text{exp}}} \right)^2 + \frac{w_f^d}{|\mathcal{D}_f|} \sum_{r \in \mathcal{D}_f} \left(\log_{10} \frac{d_r(\theta)}{s_f^d d_r^{\text{exp}}} \right)^2 \right] + J_{\text{os}}(\theta)$$

Log-normal measurement errors and uniform priors on the admissible box make minimising J exactly maximising $p(\theta | y_{\text{exp}})$, with $w_f \propto \sigma_f^{-2}$: the weights state relative credibility, they do not tune. The **log-ratio residual is symmetric under inversion** — an absolute-ratio residual penalises a factor of two high four times as hard as a factor of two low, and so rewards shrinking. The transfer factors s_f shift the 0-D targets onto the 3-D interior ($\times 1.15$ on $\langle a \rangle$ density, $\times 0.71$ on $\langle c \rangle$), so that *the calculation, not the fit*, is what agrees; J_{os} penalises non-monotonic dose response.



$\log_{10}(\text{model/exp})$	mean	s.d.	n
$\langle a \rangle$ density	-0.056	0.493	71
$\langle a \rangle$ diameter	+0.084	0.176	44
$\langle c \rangle$ density	+0.321	0.225	5
$\langle c \rangle$ diameter	-0.084	0.100	5

The calibrated set (Table 9) — † on a bound, ‡ not moved from the seed

transport — inputs from Eq. (52), not fitted

p_v	1.0200	p_i	0.7000	$p_{2i} = p_{3i}$	0.7000
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calibrated

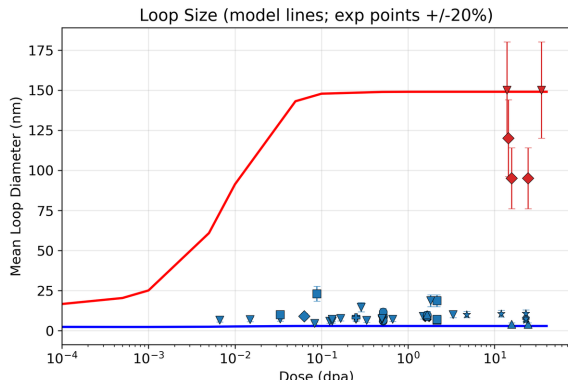
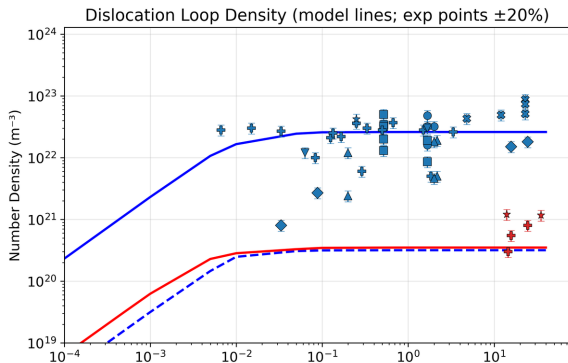
Z_v^0	1.3000†	Z_i^0	0.8935	Z_N	1.0100‡
ξ_c	2.0000†	χ	1.0000‡	E^m (eV)	0.8970
c_{LL}^a	4.7720	c_{LL}^c	1.2386	κ_{LL}	20.000†
c_{LN}^a	94.615	c_{LN}^c	1.1198	κ_{LN}	4.0000†
$m_{\text{nuc}}^{\min}$	5.8286	$m_{\text{nuc}}^{\text{vanish}}$	14.04	ν_{vanish}	0.0010†
m_a	53.79	m_c	400.00†	ε_{3i}	0.007376

- ▶ A screen over the 26 candidates found nine inert — J unmoved across their whole range — and those were held fixed
- ▶ Five bounded Nelder–Mead stages, then a joint polish: capture prefactors at fixed $p_m \rightarrow \langle a \rangle$ temperature trend $\rightarrow \langle a \rangle$ level $\rightarrow \langle c \rangle$ repair (two coalescence exponents are shared between the families) $\rightarrow$ polish
- ▶ **The order is forced by watching it fail:** fitting the $\langle c \rangle$ removal channels first commits the search to a coalescence basin it never leaves — both κ on their ceilings, basal diameter 8.6 nm — and the seed must already carry both diameters near experiment
- ▶ Four parameters rest on bounds ($Z_v^0, \xi_c, \kappa_{LL}, \kappa_{LN}$), locating the shortfall in the **capture prefactor, not the diffusion tensor**. A point estimate: the posterior has not been sampled, so no credible intervals

The 3-D march: the loops against the measurements

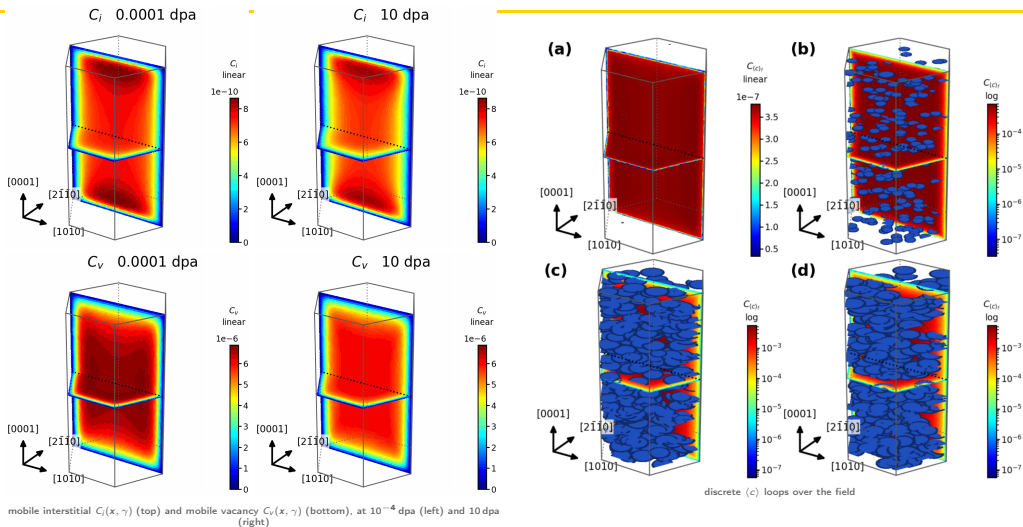
500 nm hexagonal crystal, matched self-consistent march

10^{-4} to 40 dpa at 573 K, every face a Dirichlet boundary, so both mobile fields are pinned to equilibrium there and rise to an interior plateau. **The interstitial layer is 11–34 nm and the vacancy layer far wider**, so $x = \Phi_V/\Phi_I$ — the single scalar a mean field would carry — takes a different value at every point of the grain.

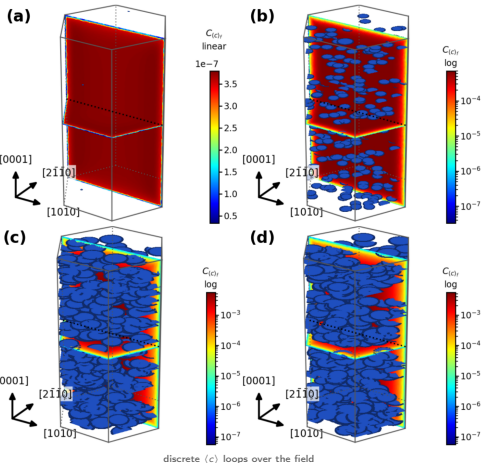


Volume-averaged loop density (left) and number-weighted mean size (right) against dose, over the TEM compilation ($\pm 20\%$). Blue: $\langle a \rangle$ interstitial (solid) and vacancy (dashed); red: $\langle c \rangle$ faulted (solid) and perfect (dashed). At 573 K and 24.5 dpa the basal comparison needs no extrapolation: $N_c/N^{\text{exp}} = 0.73$, $d_c/d^{\text{exp}} = 2.23$.

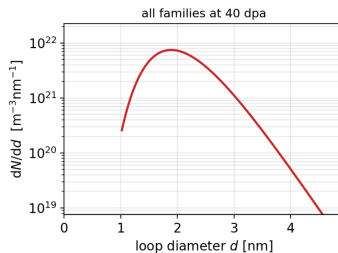
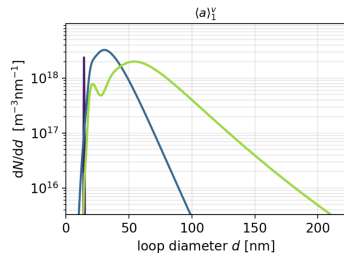
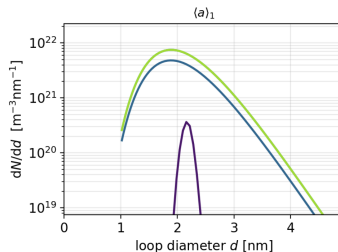
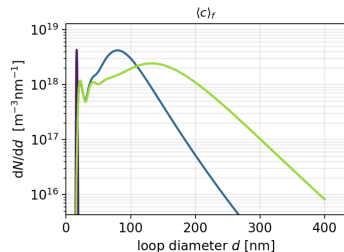
3-dimensional Defect Fields in Zirconium



The two mobile fields are shown at the ends of the march, 10^{-4} and 10 dpa at 573 K, on the 500 nm hexagonal crystal cut to show the interior. **Right:** the discrete basal $\langle c \rangle$ population the conversion draws from those fields at (a) 10^{-4} , (b) 10^{-2} , (c) 1 and (d) 40 dpa, at true radius and sampled position over the continuum content field it came from.



Reconstructed loop size distributions, α -Zr



The width is transported, not only the mean

Reconstructed size distributions over the whole crystal at 10^{-4} , 10^{-2} , 1 and 40 dpa, resolved by family.

The log-normal closure carries **three moments per family**, so the *width* of the distribution — and not only its mean — is transported.

That is what lets the continuum-to-discrete conversion **draw loop sizes from a distribution** rather than assign every loop the same size.

panels 1-3: one curve per dose

Conclusions

Advantages of RAGs

- ▶ A new material is a **graph declaration**, not a rewrite: bcc EUROFER-97 and hcp α -Zr share one core
- ▶ Forbidden reactions excluded at construction; each edge visited once — nothing double counted
- ▶ Conservation laws are left null vectors of $\mathbf{N}$ — monitored, not imposed

State-Space Size Control

- ▶ He-occupancy collapse of the gas-vacancy grid (He moment exact)
- ▶ Log bin moments with exact per-size kernels: $\sim 10^5$ size equations $\rightarrow$ a few hundred ODEs
- ▶ $308\times$ fewer equations, $43\times$ faster; d_{100} within 1% of the exact solution at 244 equations

Experimental Validation

- ▶ EUROFER-97, 330 °C, 40 dpa: **all six observables** (N , d of $\frac{1}{2}\langle 111 \rangle$, $\langle 100 \rangle$, cavities) in measured ranges
- ▶ α -Zr, 78 TEM measurements: $\langle a \rangle$ density 0.88 $\times$, diameter 1.21 $\times$; $\langle c \rangle$ diameter 100.7 vs. 95 nm
- ▶ No fitted bias factor: all biases from the diffusion tensors ($p_v=1.02$, $p_i=0.70$)

Loop Conversion Mechanisms

- ▶ $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ as inter-population edges: single-loop rotation, Marian junction ($\sim 6\%$ completion), absorption at sessile $\langle 100 \rangle$ hosts
- ▶ $\langle 100 \rangle$ share rises sigmoidally: 50% near 3 dpa, $\sim 77\%$ at 40 dpa — **kinetic, not thermodynamic**
- ▶ $\langle 100 \rangle$ family refreshed from below $\Rightarrow$ non-monotonic mean size

Co-growth Conditions

- ▶ Elastic window $1 + \varepsilon < x < Z_i^{\text{loop}}$: iron misses it by $\sim 10\times$ ($f_i^{\text{cl}} - f_v^{\text{cl}} \approx 0.44$ vs. $\lesssim 0.04$)
- ▶ Clustered SIAs deposited as loops sustain growth — **co-growth is a production-bias phenomenon**
- ▶ Cavities re-dissolve as $\langle 100 \rangle$ loops build: near-zero swelling is computed, not imposed

Spatially Resolved CD

- ▶ Frozen-mobile split: 2.1×10^8 stiff ODEs $\rightarrow \sim 7.3 \times 10^4$ independent 36-D systems, ~ 1 h on a workstation
- ▶ Resolves 8–46 nm mobile boundary layers; both atom balances close on production
- ▶ **Denuded zone** (~ 5 nm) predicted; 4–5 $\times$ higher $\langle c \rangle$ density in the interior — invisible to a mean-field k_{gb}^2

Thank you

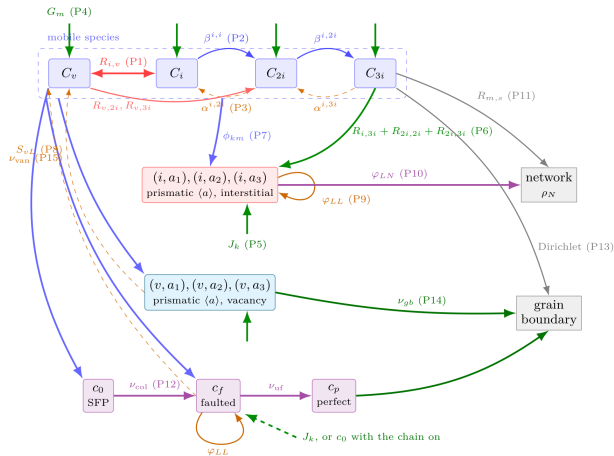
RADCLUSTER is open source.

Nasr M. Ghoniem · ghoniem@ucla.edu

Mechanical & Aerospace Engineering, UCLA

5. Backup Slides: Towards a “Digital Twin”

The Zr reaction graph and its master equation



Fifteen process classes — three with no mean-field counterpart

- ▶ **Cascade production is not one edge but two** partitions of the same displacement rate — G_m into the mobile species, J_k nucleating loops directly
- ▶ **P14 — size-gated absorption of loops at a grain boundary**, which has no meaning for a boundary that is itself only a sink strength
- ▶ **P15 — dissolution below the minimum stable size**, P14's companion gated the opposite way

The prismatic interstitial family also nucleates *homogeneously* from $2i+2i$, $i+3i$ and $2i+3i$, so **the mobile ladder and the loop populations are not separable even in principle.**

Reaction admissibility graph for the nine families, carrying all fifteen process classes.

Cascade production is drawn as a *partition* of one displacement rate, not a transfer

Verification against the exact discrete solution

$I = V = 4000$; the $\frac{1}{2}\langle 111 \rangle$ coarsening channel is disabled in both arms so the two solvers integrate identically the same equations; scored at 2.0000 dpa. Densities in 10^{21} m^{-3} (loops) and 10^{20} m^{-3} (cavities), diameters in nm; parentheses are percent deviation from D.

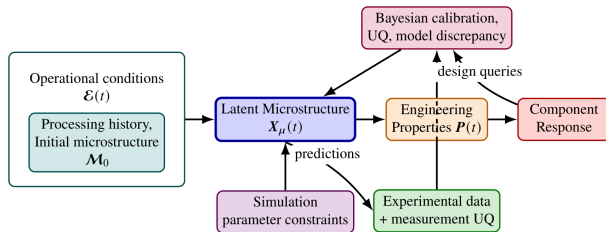
rung	N_{eq}	total	N_{111}	N_{100}	d_{100}	N_{cav}	d_{cav}
D (exact)	8006	12006	12.568	1.065	5.741	2.948	2.544
C1 (1/10)	830	1242	12.335 (-1.9)	1.044 (-2.0)	5.690 (-0.9)	2.555 (-13.3)	2.273 (-10.6)
C2 (1/40)	244	364	12.144 (-3.4)	1.044 (-2.0)	5.681 (-1.0)	2.574 (-12.7)	2.279 (-10.4)
C3 (1/160)	108	161	7.027 (-44.1)	1.018 (-4.4)	5.349 (-6.8)	3.219 (+9.2)	2.451 (-3.6)

Solver configuration and cost

configuration	N_{eq}	wall clock
<i>exact discrete arm</i>		
$I = V = 1000$	2006	5.2 min to 20 dpa
$I = V = 2000$	4006	61 min $\rightarrow$ 0.038 dpa
$I = V = 10^4$	—	diverges in cost
<i>production domain, $I = V = 10^4$</i>		
GMRES+Woodbury (ref.)	20007	9198 s
bin-moment, hat $P=2$	65	213 s
GMRES + Jacobi	—	times out
lognormal $P=3$	—	times out
sliding window	—	times out

- ▶ Common settings: EUROFER-97 host, fission spectrum, $G = 10^{-7} \text{ dpa s}^{-1}$, 330 °C, absolute tolerance 10^{-20} , $i_{\text{mob}} = v_{\text{mob}} = 5$
- ▶ **Jacobian**: bordered-banded, inverted by **Sherman–Morrison–Woodbury preconditioned GMRES**; the same problem with Jacobi preconditioning times out
- ▶ The exact arm is confined to small domains by the serial $O(N_c^2)$ coarsening pair sum — a hundredfold larger at $I = 10^4$ than at 10^3
- ▶ **d_{100} holds to 1% from 830 equations down to 244** — a 30 $\times$ cost reduction for a one-percent error; N_{100} to 2%
- ▶ N_{111} breaks at C3: once i_{disc} falls to 25 the peak of the $\frac{1}{2}\langle 111 \rangle$ distribution lies inside the binned region
- ▶ $\text{rto1 } 10^{-4}$ versus 10^{-6} moves no observable in any reported digit — the tolerance is not a hidden knob

Cluster dynamics as the latent state



A physics-informed, uncertainty-quantifiable microstructure-to-properties architecture. The physics core is the evolving microstructural state, not the surrogate.

$$\mathcal{E}(t)[\mathcal{M}_0] \longrightarrow X_{\mu}(t) \longrightarrow P(t)$$

operational history $\rightarrow$ latent microstructure $\rightarrow$ strength, toughness, creep, conductivity

"The microstructure is treated as the latent state that connects the operational environment to properties."

Where cluster dynamics sits

- ▶ DFT / MD cascade statistics supply *bounded priors*, not point values
- ▶ RADCLUSTER is the bulk irradiation engine; SRCD/CP carries the spatially resolved fields; DisloCluster supplies bias factors and sink strengths that *replace fitted values*
- ▶ Microstructure-to-property closures are mechanism-specific rather than empirical, e.g. dispersed-barrier hardening over the *predicted* obstacle spectrum, $\Delta\sigma_y = M\mu b \left[\sum_k \alpha_k^2 N_k d_k \right]^{1/2}$
- ▶ Component response: nonlinear cyclic thermo-structural FE against design limits

The credential this talk offers

The Zr calculation — $\sim 10^7$ ODEs, 0 to 100 dpa, **nine microstructural components, no adjustable macroscopic strain parameter** — is the validation case being transferred to bcc EUROFER97 / F82H.

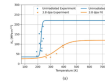
...and the uncertainty that makes it a twin

$$p(\mathbf{P} | \mathcal{M}, \mathcal{E}(t), \mathcal{D}) = \int \underbrace{p(\mathbf{P} | \mathbf{X}_\mu, \theta)}_{\text{closure}} \underbrace{p(\mathbf{X}_\mu | \mathcal{M}, \mathcal{E}, \theta)}_{\text{kinetic — this is CD}} \underbrace{p(\theta | \mathcal{D})}_{\text{posterior}} d\mathbf{X}_\mu d\theta$$

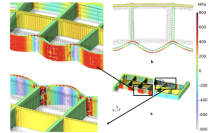
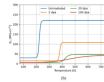
- Migration and binding energies, capture radii, sink strengths and bias factors are carried as *distributions rather than single constants*
- An explicit **model-discrepancy** term, so parameter uncertainty is not silently inflated to absorb structural error
- $\text{Var}(Y) = V_{\text{source}} + V_{\text{kinetics}} + V_{\text{micro}} + V_{\text{constitutive}} + V_{\text{discrepancy}} + V_{\text{meas}} + \dots$
- Sobol' indices and **expected information gain** rank candidate experiments — **in the currency of component life, not parameter variance**

Four-level validation, with an anti-circularity rule

I parameters → II microstructure → **III properties (the migration gate)** → IV extrapolation. Calibration data are *never* reused as validation evidence.



(a)



(b)

(a) Fracture-toughness predictions for irradiated F82H; (b) FNSF first-wall / blanket thermo-structural analysis.

The payoff, stated as a target

A fully nonlinear cyclic analysis of the FNSF first wall in F82H predicts **roughly a one-year life at 1.2 MW/m² neutron wall loading and 0.25 MW/m² surface heat flux** — limited by flow localization, thermal creep damage and fast fracture from radiation-induced toughness loss, *not* by high-fluence swelling. The object is to replace that *deterministic* number with a **failure-probability history with stated confidence**.

Design-relevant target 100 dpa, extrapolating to 100–200 dpa — while most fission data end near 20 dpa.